

Ancient Anorthosites on Mars: Widespread in the Lower Crust

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Abstract

We report extensive detections of spectrally feldspathic, possibly anorthositic, lithologies within outcrops uplifted by the Argyre basin-forming impact on Mars. The radial position of spectrally feldspathic outcrops from the basin center indicates that they are stratigraphically above olivine-bearing material and below low calcium pyroxene-bearing rocks, and the inferred depths suggest that spectrally feldspathic lithologies are present in the lower martian crust in the circum-Argyre region. This stratigraphy bears striking resemblance to that observed at the Hellas basin rim and indicates that feldspar-rich rocks could be a pervasive component of the lower crust across the entire southern highlands of Mars. Recent evidence from geophysical data suggest a lower-density lower crust leading to hypotheses that the lower martian crust could be dominated by feldspar. However, plagioclase-rich outcrops constitute only $\sim 4\%$ of the mapped pre-Argyre crust, which is insufficient to fully account for the lower density inferred for the lower crust. Importantly, these spectrally feldspathic rocks are present alongside pyroxene- and olivine-bearing units, a suite of compositions commonly associated with layered mafic intrusions (LMI). We suggest that a process early in Mars' history produced large volumes of melt that cooled slowly to form LMI-like complexes and that these complexes represent an aspect of primary martian crust. The melt-producing process ostensibly affected the whole southern hemisphere, and possibly the globe. These observations offer direct evidence to support and refine existing models of early crust formation, addressing key questions regarding crustal density variations and formation mechanisms that have previously lacked widespread observational constraints.

Main

In contrast to Earth and Venus, Mars preserves a surface rock record dating back to ~ 4.55 Ga (Humayun et al., 2013; Kruijer et al., 2020). The preservation and exposure of ancient rocks (≥ 4.1 Ga, pre-Noachian) at the surface provides a unique opportunity to study early crustal formation through remote sensing and surface investigations. Understanding how and when evolved rocks first formed on terrestrial planets remains a major open question in planetary geology. Since the processes that formed Mars' primary crust share more similarities with Earth- and Venus-sized bodies than with lunar-sized bodies (Elkins-Tanton, 2012), studying the oldest Martian crust provides critical insights into early crustal evolution with potential for comparative planetology.

Noteworthy among the ancient rocks of Mars are compositions apparently dominated by plagioclase feldspar (Fig. 1). Ancient feldspathic and silicic rocks measured *in situ* provide clear evidence that crustal differentiation (i.e., within-crust differentiation as distinct from the differentiation of the crust from the mantle) occurred early in Mars' history (e.g., Stolper et al., 2013; Sautter et al., 2015); though the formation mechanism for these rocks was likely fractional crystallization of basaltic magmas (Udry et al., 2018; Ostwald et al., 2022) rather than a plate-tectonic-style process (Sautter et al., 2015, 2016) akin to the formation of continental crust on Earth. Whether such differentiation occurred only at isolated locations or globally is an open question. Here, we present evidence for a widespread, plagioclase-rich

component to the lower martian crust that is likely buried throughout the entire southern highlands. Our results suggest a hemispheric, if not global, process that resulted in differentiated magmas over ~ 4.1 billion years ago on Mars.

Though increasingly recognized as important constituents of the Martian crust, spectrally feldspathic rocks are relatively rare compared to basaltic materials based on orbital observations (Ehlmann and Edwards, 2014; McSween, 2015; McSween et al., 2009, 2023). Many detections of plagioclase-rich compositions come from isolated exposures in Noachian and older terrains in the southern highlands, interpreted as possible intrusive bodies (Quantin et al., 2012; Carter and Poulet, 2013; Wray et al., 2013; Sautter et al., 2016; Viviano-Beck et al., 2017; Payré et al., 2022). Other plagioclase-rich occurrences are linked to likely volcanic units found in Hesperian-aged terrains, such as Syrtis Major (Wray et al., 2013; Eggers et al., 2021; Rogers and Farrand, 2022), western Arcadia Planitia (Farrand et al., 2021), and the Eridania basin (Michalski et al., 2023).

Other studies have revealed more extensive occurrences of plagioclase-rich outcrops. Phillips et al. (2022) identified widespread plagioclase-rich outcrops across the northern rim of the Hellas basin. Viviano-Beck et al. (2017) initially reported, and Flahaut et al. (2023) corroborated, the presence of extensive plagioclase-rich compositions in the lower walls of Valles Marineris. The Valles Marineris exposures are plausibly Noachian or older, and are interpreted to be intrusive bodies such as sills or other hypabyssal intrusions (Viviano-Beck et al., 2017), although some evidence may suggest they could alternatively be effusive volcanic rocks (Flahaut et al., 2023). Additionally, nine feldspar-rich sites were identified in the Terra Sirenum/Cimmeria region by Payré et al. (2022), an area with anomalous geochemical and magnetic signatures and suggested to be an ancient crustal block (Bouley et al., 2020). Recent findings by the Perseverance rover at Jezero crater have also revealed feldspar-dominated and spinel-bearing rocks (Bedford et al., 2024; Schmidt et al., 2025; Mansbach et al., 2024; Lawson et al., 2025) that could represent material uplifted by the Isidis-forming impact, providing additional evidence for uplifted differentiated crustal materials. Therefore, though plagioclase-rich rocks are found across the southern highlands of Mars, it appears that a diversity of petrogenetic processes including plutonic and volcanic activity, and possibly impact-related processes (Grieve et al., 1991; Hurwitz and Kring, 2014; Vaughan and Head, 2014; Koeppel et al., 2020) have contributed to their formation from the pre-Noachian through the early Hesperian Epochs. This work focuses on the most ancient (pre-Noachian) plagioclase-rich outcrops.

The lower crust of Mars has been hypothesized to be feldspar-rich to explain its apparent low density based on geochemical and geophysical data (Baratoux et al., 2014; Goossens et al., 2017; Knapmeyer-Endrun et al., 2021). Despite the growing number of observations of plagioclase-rich rocks, it remains unclear from surface remote-sensing observations whether a lower-crust plagioclase-rich component is present globally. Here, we report results from an investigation into outcrops uplifted from depth by the Argyre basin-forming impact to test the hypothesis that the martian crust contains a global feldspathic component. Following the identification of abundant plagioclase-rich outcrops along the northern Hellas rim (Phillips et al., 2022), we find that plagioclase-rich outcrops are also ubiquitous in Argyre-uplift materials and share many similarities with those discovered along the northern Hellas rim,

supporting the hypothesis that these materials are widespread in the lower martian crust.

Plagioclase-rich outcrops sandwiched between mantle and upper crust

We investigated the composition of uplifted material surrounding Argyre using ~ 180 m/pixel, 72-band multispectral mapping data (MRDRv4, Seelos et al. 2023) from the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM, Murchie et al., 2007) instrument onboard the Mars Reconnaissance Orbiter (Zurek and Smrekar, 2007). Reflectance spectroscopy is a useful tool for identifying many silicate minerals, including major rock-forming minerals such as olivine, pyroxene, Fe-bearing plagioclase, and oxides (Adams and Filice, 1967; Adams, 1968; Clark and Roush, 1984; Burns, 1993).

We examined 40 CRISM MRDR tiles (Extended Data Table 1) for the presence of primary rock forming silicate minerals. Of the 1349 outcrops identified to contain signatures of primary rock forming minerals, 476 showed absorption features consistent with olivine + high-calcium pyroxene (HCP), 140 were consistent with type 1 olivine, 320 with type 2 olivine, 282 with low-calcium pyroxene (LCP), and 131 were consistent with plagioclase (Fig. 2; see Methods for details on spectral categorization). Type 1 olivine is interpreted as more Mg-rich or smaller grained and type 2 olivine is more Fe-rich or larger grained (Viviano et al., 2014; Ody et al., 2013).

The emplacement mechanism of uplifted massifs resulting from basin-forming impacts, such as the one that formed Argyre, is primarily attributed to structural uplift (Tanaka and Leonard, 1995; Johnson et al., 2016) and to a lesser extent ejected blocks (Johnson et al., 2016). Structural uplift refers to the combined effects of collapse of a large central uplift – analogous to the formation of peak ring structures in complex craters (Johnson et al., 2016) – and isostatic rebound and crustal thinning from transient crater formation (Tanaka and Leonard, 1995), which leads to an extensional tectonic regime that causes normal faulting. The relative stratigraphy of material uplifted this way would be preserved (with some disruption) as opposed to the inverted stratigraphy expected for an overturned flap (Kenkmann et al., 2014). Ejected blocks of material would also maintain the trend of deeper material emplaced closer to the center of a basin, although more mixing of materials originating from different depths would occur than for structurally uplifted massifs (Extended Data Fig. 1, Trowbridge et al. *submitted*). Uplifted materials closer to the basin center would thus tend to be stratigraphically below those further from the basin center. However, the complex emplacement processes associated with basin-forming impacts would lead to some local disruption of the original stratigraphy, and the potentially oblique impact that formed the Argyre basin would lead to asymmetric emplacement of ejected blocks (Extended Data Figs. 1 and 2). The key assumptions necessary for our analysis, which are supported by modeling, are that Argyre excavated to the mantle, and that material closer to the basin center originates from greater depths than material further from the basin center on average. In this work, the distance of each outcrop from the basin center is used as a proxy for pre-uplift stratigraphy (Fig. 2).

Olivine-bearing units (type 1, type 2, and olivine + HCP) vastly dominate the compositions exposed within approximately 720 km of the Argyre basin center (Fig. 2). It is likely that some of these olivine-bearing compositions represent uplifted mantle material (Extended Data Fig. 1; Koeppen and Hamilton, 2008; Ody et al., 2013). Among the olivine-bearing categories, olivine + HCP outcrops are concentrated lowest along the rim (based on area-weighted average distance calculations, Extended Data Fig. 3) where possible mantle material is expected, raising the possibility of wehrlite/lherzolite in the upper martian mantle (McGetchin and Smith, 1978; Treiman, 2025; Bernadet et al., 2025). However, the presence of absorption features consistent with olivine and HCP (augite) in the CRISM spectra is not conclusive evidence that these outcrops are wehrlitic or lherzolitic and could have alternative interpretations (Extended Data Fig. 4). Beyond ~ 720 km, LCP is the most abundant mineral category. The position of LCP-dominated compositions is suggestive of upper pre-Noachian crust (Buczkowski et al., 2010; McNeil et al., 2025). The material between these two compositions is interpreted as a comminuted mixture of mantle, lower-, and upper-pre-Noachian crust.

The plagioclase detections described here are concentrated between ~ 330 and 890 km from the basin center and are absent beyond 890 km. The highest relative proportion of plagioclase ($\sim 29\%$) occurs between 430 and 470 km from the basin center. The overall pattern revealed by the distribution of mineral categories with radial distance (Fig. 2) suggests a clear transition from compositions dominated by olivine at lower stratigraphic positions to those dominated by LCP in the upper stratigraphic positions (Extended Data Fig. 2, 3). This transition occurs between ~ 450 and 810 km distance from the Argyre basin center. Plagioclase-rich rocks appear to be positioned lower in the crust alongside the stratigraphically lower olivine-bearing rocks, some of which may represent mantle materials, and below the stratigraphically higher LCP-bearing rocks (Fig. 2, Extended Data Fig. 2, 3). The gross stratigraphy observed in the circum-Argyre uplifted materials is consistent with the stratigraphy observed in the Hellas basin rim (Phillips et al., 2022) and with the well-preserved stratigraphic relationships exposed in the lower walls of Valles Marineris (Viviano-Beck et al., 2017). The similarity of the observed stratigraphy at Argyre to the intact stratigraphy in Valles Marineris lends support to the interpretation of stratigraphic relationships in the circum-Argyre region, despite the complex emplacement processes associated with the basin-forming impact.

Local contexts, visual appearance and color analysis

Three general categories were identified for outcrops when considering their local geomorphological context: massifs, crater rims, and valley walls. Circum-Argyre massifs are the principal target of interest for our study because they directly represent pre-Argyre rocks excavated by the impact process. We include crater-rim outcrops in our analysis because impacts on the Argyre rim could uplift material that was originally uplifted by Argyre. Exposures within valley walls are also of interest because valleys cut into the circum-Argyre material and thereby expose buried rock.

Plagioclase-rich outcrops on massifs typically appear as light-toned, ridge-like capping units

overlying other mineral detections even within the same massifs (Extended Data Figs. 5 and 6). Many plagioclase-rich outcrops exhibit massive, homogeneous light-toned appearances that resemble terrestrial massif anorthosites or thick anorthositic members of layered mafic intrusions (Fig. 1, Extended Data Fig. 5; Ashwal 1993). Massive textured outcrops also appear as rounded or angular blocks (Extended Data Fig. 5) reminiscent of the “mega breccia” reported in the circum-Isidis region by Scheller and Ehlmann (2020); however, some of the “mega breccia” along the Argyre rim are larger in scale (up to 433 meters at Isidis versus several kilometers in scale at Argyre). Thermal Emission Imaging System (THEMIS; Christensen et al. 2004) thermal inertia (Ferguson et al., 2006) and nighttime infrared data (Edwards et al., 2011) over massif outcrops indicate high thermal inertia values relative to surrounding units, suggesting they are well-consolidated exposures rather than loose, unconsolidated materials.

The spatial relationships of neighboring outcrops (defined here as outcrops within 5-km distance of one another) were assessed using a graph network analysis to understand patterns in local associations and relative elevations (Extended Data Fig. 6). The most common neighbor of plagioclase-bearing outcrops (aside from other plagioclase-bearing outcrops) are type 2 olivine and LCP (23 and 22 counts each), followed by type 1 olivine (12 counts), and olivine + HCP (9 counts). The proximal relationships indicate a spatial association among plagioclase, type 2 olivine and LCP (that is, these compositions commonly appear on or near the same massifs) and that plagioclase is locally higher in elevation than its mafic neighbors (Extended Data Fig. 6). This may evidence local (kilometer-scale) stratigraphic relationships that are distinct from the basin-wide distribution of minerals that likely represents the overall pre-Argyre crustal stratigraphy.

The plagioclase-bearing outcrops associated with craters that we identified in this study are found exclusively on crater rims, not within crater floors, suggesting they represent uplifted material rather than later volcanic or sedimentary infill or local differentiated impact melt or breccias. Valley walls primarily expose material in the outer regions of the Argyre rim. Though LCP dominates these exposures, plagioclase and olivine are detected in valleys lower down on the rim, consistent with the basin-wide stratigraphic relationship observed in the region.

Analysis of High Resolution Imaging Science Experiment (HiRISE; McEwen et al., 2010) infrared, red, blue-green (IRB) color data reveal that plagioclase-rich outcrops consistently appear as higher reflectance, light-toned materials relative to surrounding terrain (Fig. 1, Extended Data Figs. 5, 7 and 8). Dark subtraction methods (Tornabene et al., 2018) were applied to three HiRISE images (ESP_020799_1385, ESP_031269_1395, and ESP_069064_1390) to remove additive atmospheric residuals in the data and assess the colors associated with CRISM-identified compositions. Outcrops categorized as plagioclase with CRISM show lower BG/IR and Red/IR band ratio values than neighboring mafic outcrops (Extended Data Fig. 7), attributable to an absorption feature present with LCP and olivine but not present with plagioclase that overlaps with the HiRISE IR channel (centered at ~ 900 nm). Light-toned outcrops that were either too small to be resolved by CRISM MRDR data or not covered by the data were also investigated with HiRISE color parameters and show the same relative trend among light- and dark-toned material within the same scene (Extended Data Figs. 7

and 8), which suggests that the CRISM-based detections of plagioclase-rich outcrops are a minimum estimate of the full distribution of such outcrops in the circum-Argyre region.

Possible lithology of plagioclase-bearing outcrops

The plagioclase-rich outcrops are characterized by a broad absorption feature centered around 1.25 μm , indicative of Fe^{2+} -bearing plagioclase feldspar (Adams and Filice, 1967; Adams et al., 1973; Adams and Goullaud, 1978). This spectral signature is observed if plagioclase is the dominant mineral visible at the surface; small amounts ($\sim 15\%$) of mafic minerals like pyroxene or olivine can mask the plagioclase absorption feature (Crown and Pieters, 1987; Pompilio et al., 2007; Serventi et al., 2013, 2015; Cheek and Pieters, 2014). However, specific, grain-size dependent, combinations of plagioclase with other minerals may also yield a reflectance spectrum dominated by the 1.25- μm absorption feature of plagioclase (“spectrally feldspathic” rocks), which means the 1.25- μm feature of Fe-plagioclase is not sufficient to classify the lithology of the host rock (Rogers and Nekvasil, 2015; Barthez et al., 2023; Vannier et al., 2024). Supplementary Note 1 of Phillips et al. (2022) presents a full discussion of this topic. Two key points are relevant to this study: First, Rogers and Nekvasil (2015) demonstrated that volcanic rocks with $\sim 50\%$ large plagioclase phenocrysts and $\sim 50\%$ pyroxene by volume can produce plagioclase-dominated reflectance spectra if the pyroxene grain size is sufficiently small. Second, felsic rocks such as granite can also exhibit reflectance spectra dominated by the 1.25- μm plagioclase absorption feature.

THEMIS thermal infrared data provide additional constraints on the composition of spectrally feldspathic outcrops. We assessed nine plagioclase outcrops across four THEMIS scenes (Extended Data Table 2; see Methods for details on processing) as well as 23 type 1 olivine, 15 type 2 olivine, 50 HCP + olivine, and 19 LCP outcrops within the same THEMIS scenes. Two metrics that track the silica content of target lithologies were calculated: the weighted absorption center (WAC; Smith et al. 2013; Amador and Bandfield 2016) and cubic spline fit minimum (CSFM; Rogers and Nekvasil 2015). Plagioclase-bearing outcrops show WAC values ranging from $\sim 9.9 - 10.2 \mu\text{m}$ with a 10.1 μm median and CSFM values ranging from $\sim 9.3 - 11.0$ with a median of $\sim 9.5 \mu\text{m}$. Plotting WAC versus CSFM further elucidates the spectral character of the outcrops. Fig. 3 shows WAC vs. CSFM for THEMIS observations compared to reference spectral library data of felsic, intermediate, mafic, ultramafic, anorthositic and gabbroic rocks. The THEMIS data for plagioclase-bearing outcrops align with intermediate, gabbroic, and anorthositic compositions, but not with felsic compositions.

The reflectance and thermal infrared spectral characteristics of plagioclase-rich outcrops, their close spatial association with pyroxene and olivine outcrops at $\leq 5\text{-km}$ spatial scales, higher relative position compared to olivine and pyroxene in local contexts on average, their light-toned appearance and high reflectance compared to mafic compositions, and the dark and light-toned layering observed within individual plagioclase-bearing massifs are consistent with a differentiated magmatic complex. Evidence for a volcanic origin, such as columnar jointing or flow structures, is not observed. The preferred interpretation of these rocks is that they represent layered mafic intrusion (LMI) style complexes (Fig. 4). Anorthositic layers in LMIs form through fractional crystallization and flotation of plagioclase in a cool-

ing, typically hypabyssal, mafic magma body (Ashwal, 1993). The modest percentage of plagioclase-rich outcrops at any radial distance from Argyre (interquartile range = 1.3 – 7.1% of total detections at distances where plagioclase is present) is consistent with the typical proportion of anorthositic rocks in terrestrial LMIs (Ashwal, 1993, 2021). However, we cannot completely rule out contributions from previous impact-basin melt bodies, which may also undergo differentiation, concentrate minerals and form cumulate rocks akin to LMI.

Alternatively, the light-toned, massive appearance of many plagioclase-rich outcrops in the Argyre region could be interpreted as evidence for massif-type anorthosites (Ashwal, 1993; Arndt, 2013; Keller et al., 2024). Massif-type anorthosites on Earth are distinct from anorthosite formation as layers in LMIs. Terrestrial massif-type anorthosites form large, relatively homogeneous bodies that can extend over thousands of square kilometers, whereas anorthositic components of LMIs typically form discrete layers that can be hundreds of kilometers in extent (e.g., Bushveld Complex, Latypov et al. 2020) and kilometers thick (e.g., Mineral Lake intrusion, Wisconsin; Ashwal 1993). Hypotheses for massif-type anorthosite formation involve an elevated crustal temperature gradient, extensive, long-lived mafic magmas, and density-driven extraction and ascent of plagioclase crystal mushes (Emslie et al., 1994; Keller et al., 2024). The tectonic setting for massif-type anorthosites is rifting at a prior convergent zone, a setting unlikely for Mars, and the associated rock types are mangerite (feldspar and orthopyroxene), charnockite (quartz, feldspar, and orthopyroxene), and granitoids (Emslie et al., 1994; Hamilton et al., 2010; Arndt, 2013; Keller et al., 2024). This suite of rocks, referred to as AMCG, are not observed in the circum-Argyre region, further leading us to favor differentiation from basaltic melts into LMI-style intrusions rather than a massif-type anorthosite interpretation for the observations.

Discussion of Early Crustal Formation Processes

The observed mineralogical distribution around the Argyre basin provides insight into the nature and origin of Mars’ earliest crust. The striking similarities between plagioclase-rich outcrops exposed at both Argyre and Hellas basins—separated by over 4,000 km—suggest they represent the same component of the lower Martian crust. Spectral, morphological, and contextual characteristics of these outcrops are consistent across both regions, including their association with olivine and pyroxene-bearing units, their light-toned appearance, presence of both layered and massive textures, and their relative stratigraphic position along the basin rims (Phillips et al., 2022). The spatial extent indicates these compositions are prevalent beneath much of the southern highlands, if not globally. The creation of such a widespread and compositionally consistent crustal component implies vast volumes of magma derived from a similar source. We favor two scenarios that could explain the generation of such enormous melt volumes across the southern hemisphere: 1) overturn of a post-magma-ocean cumulate pile (Borg and Draper, 2003; Elkins-Tanton et al., 2005); 2) melt and magma generation from basin-forming and/or frequent impacts (Koeppel et al., 2020; Manske et al., 2021; Black and Marchi, 2024). A third alternative could be partial melting of the southern hemisphere’s lower crust (Bonnet Gibet et al., 2025; Bernadet et al., 2025).

Magma Ocean and Mantle Processes.

A whole mantle magma ocean would have formed a gravitationally unstable cumulate mantle with less dense Mg-rich cumulates below more dense Fe-rich cumulates (Elkins-Tanton et al., 2005). Subsequent mantle overturn would have generated large volumes of melt through decompression. These melts would have ascended and cooled slowly in the crust, consistent with formation of differentiated igneous complexes through fractional crystallization. A degree-one overturn geometry (Borg and Draper, 2003; Elkins-Tanton et al., 2005; Debaille et al., 2008) would have resulted in similar crustal compositions across the southern hemisphere, which more easily explains our observations than a higher degree overturn geometry (Scheinberg et al., 2014) that would have produced spatially variable crustal compositions across the southern highlands.

Impact Melt Differentiation.

A second hypothesis involves the differentiation of a single massive, or many overlapping, impact melt sheet(s). The Borealis basin-forming impact, which is a leading hypothesis for the creation of Mars' crustal dichotomy (Don and Steven, 1984; Andrews-Hanna et al., 2008; Marinova et al., 2008; Nimmo et al., 2008), would have generated an enormous volume of melt. Differentiation of this melt sheet could have produced anorthositic rocks through plagioclase flotation (Koeppel et al., 2020; Manske et al., 2021), later uplifted by the Argyre and Hellas impacts. This scenario is particularly intriguing given that the impact melt is expected to concentrate primarily within the impact basin (Marinova et al., 2008; Ballantyne et al., 2023) and the feldspathic outcrops are found in the southern hemisphere. This raises the possibility that if a large impact formed the crustal dichotomy, it may have impacted the southern hemisphere rather than the northern hemisphere (Reese et al., 2010; Golabek et al., 2011; Leone et al., 2014; Ballantyne et al., 2023; Cheng et al., 2024), creating a localized magma ocean in the southern hemisphere.

The cumulative effect of high frequency bombardment of early Mars would likely have created massive volumes of melt (both impact melt and crustal partial melts) even in the absence of a single, large, dichotomy-forming impact (Koeppel et al., 2020; Manske et al., 2021; Black and Marchi, 2024). The expected fate of such melts is ascent to a depth of neutral buoyancy in the crust, contributing to a compositional stratification of the early crust (Black and Marchi, 2024). Additionally, differentiation of surface melts is known to occur at terrestrial impact sites, leading to kilometers-thick fractionated melt sheets (Therriault et al., 2002; O'Connell-Cooper and Spray, 2011). With the high impact flux and high thermal gradient in the pre-Noachian, it is plausible that this process of compositional layering at individual impact sites could have occurred repeatedly, producing a compositionally stratified crust (Koeppel et al., 2020).

Crustal Melting.

A third possibility is that evolved compositions formed through partial melting of hydrothermally altered mafic crust (Bonnet Gibet et al., 2025; Bernadet et al., 2025). We disfavor this scenario as an explanation for the pre-Noachian feldspathic crust because silicic (i.e., felsic) rocks would be the expected product of the partial melting—compositions not observed in this study.

Two plausible scenarios for crustal melting have recently been proposed. The first involves crustal thickness differences between Martian hemispheres. In the southern highlands, where crust is thicker, higher concentrations of heat-producing elements create steeper temperature gradients. This thermal insulation, combined with thinner lithosphere and higher mantle melt fractions beneath these regions, enhances temperatures at the crustal base, promoting partial melting at depths up to approximately 15 km (Bonnet Gibet et al., 2022, 2025). The resulting evolved magmas would ascend, potentially undergo assimilation and fractional crystallization, and crystallize at various crustal depths—a process sharing characteristics with silicic rock formation in Iceland (Jónasson, 2007; Martin and Sigmarsson, 2007, 2010).

The second mechanism involves interaction between basaltic melts and serpentized primordial crust (Bernadet et al., 2025), plausible for early Mars given evidence for widespread alteration (Bibring et al., 2006; Bishop et al., 2008; Ehlmann et al., 2009; Ehlmann and Edwards, 2014; Wordsworth et al., 2017; Viviano et al., 2023; Carter et al., 2023; Royer et al., 2024) and the possibility of a dense steam atmosphere persisting for several million years post-accretion (Abe, 2011; Scherf and Lammer, 2020). In this model, water both enables serpentinization of ultramafic protocrust and significantly lowers rock melting temperatures. Thermodynamic modeling indicates a minimum water content of ~ 100 ppm in the bulk silicate Mars would be necessary (Bernadet et al., 2025). Through shallow (< 15 km) fluid-assisted reactions between serpentinite and basaltic melts, melts with felsic compositions could form. Models suggest this process might generate up to $\sim 40\%$ of Martian felsic crust, particularly during two periods: very early Martian history (4.4–4.5 Ga) and during the Hesperian Epoch (Bernadet et al., 2025).

However, the spectral characteristics of our detections are more consistent with intermediate/gabbroic/anorthositic rocks than with silicic/felsic compositions. Though partial melting of possibly hydrated crust may explain some Hesperian-aged spectrally feldspathic detections (Farrand et al., 2021; Rogers and Farrand, 2022; Michalski et al., 2023), it seems less applicable to the pre-Noachian compositions presented here. For the few silicic rocks encountered *in situ* on Mars, fractional crystallization of basaltic magmas may better explain their formation than hydrous crustal melting (Christensen et al., 2005; Udry et al., 2018; Ostwald et al., 2022).

The presence of Fe^{2+} in plagioclase, which enables its spectral identification, provides additional petrogenetic information. Fe^{2+} incorporation into plagioclase is enhanced under reducing conditions, with high melt polymerization (viz., higher SiO_2), low water activity, high $\text{Fe}^{\text{Tot}}/\text{Al}_2\text{O}_3$, and in high temperature melts that rapidly ascend to cool at low pressures (Wenk and Wilde, 1973; Longhi et al., 1976; Smith and Brown, 1988; Phinney, 1992; Ashwal, 1993; Lundgaard and Tegner, 2004). Though the presence of Fe^{2+} in plagioclase

is not diagnostic of the petrogenetic environment, favorable conditions for Fe^{2+} substitution into plagioclase include dry, high temperature, rapidly ascending melts that undergo fractional crystallization at low pressures and allow for the extraction of plagioclase as flotation layers. Such conditions are more consistent with either a mantle overturn scenario or impact-generated melts than with partial melting of hydrated crust.

The above mechanisms need not be mutually exclusive. The early Martian crust likely formed through a complex interplay of processes operating at different spatial and temporal scales. Cumulate mantle overturn may have produced voluminous melts that solidified into differentiated igneous complexes while concurrent impact events generated substantial near-surface melt across the planet.

In the elevated geothermal and water-rich environment of early Mars, surface volcanics and shallow crustal materials would have undergone hydrothermal alteration (Viviano and Phillips, 2019; Wu et al., 2025). After subsequent burial and diagenesis, these altered materials could have experienced partial melting, producing intermediate to evolved compositions. The spatial distribution of these evolved compositions would likely be heterogeneous, with concentrations in regions of thicker crust or areas with significant hydrothermal activity. Such a multi-stage petrogenetic history is likely necessary to explain the diverse suite of spectrally feldspathic materials observed across Mars (Carter and Poulet, 2013; Wray et al., 2013; Rogers and Nekvasil, 2015; Viviano-Beck et al., 2017; Farrand et al., 2021; Payré et al., 2022, 2024; Phillips et al., 2022; Flahaut et al., 2023; Michalski et al., 2023).

Implications for crustal formation, density, and remnant magnetism

The presence of plagioclase-rich rocks and associated mafic lithologies in ancient crustal materials uplifted by the Argyre impact has implications for understanding Mars' early crustal evolution. The similar detection from orbit of plagioclase-rich materials at both Argyre and Hellas (Phillips et al., 2022) and in the circum-Isidis region by the Perseverance rover at Jezero crater suggests this is a hemispheric, if not global, component of Mars' lower crust. These observations support models of early Martian evolution involving hemisphere-scale or global-scale production of magma, though the specific mechanism remains uncertain. Additionally, these results show crustal differentiation can occur without plate tectonics – possibly via fractional crystallization of large mafic magma bodies – a process which may be applicable to early Earth (Kranendonk, 2010).

Though plagioclase-rich rocks are widespread in the circum-Argyre region (131 unique outcroppings across the basin), they constitute only about 4% by area of the pre-Argyre crust mapped with CRISM data in this study. This finding is crucial for understanding the lower martian crust because it directly addresses the hypothesis that plagioclase-rich rocks may explain the overall lower density of Mars' lower crust inferred from geochemical and geophysical data (Baratoux et al., 2014; Goossens et al., 2017; Knapmeyer-Endrun et al., 2021; Drilleau et al., 2023; Joshi et al., 2023). Baratoux et al. (2014) proposed a lower density component in the range of 2.5 – 2.9 g/cc, consistent with more recent estimates informed by InSight (e.g., Knapmeyer-Endrun et al., 2021). If the ~4% proportion of plagioclase-rich rocks observed in this study is representative of the true abundance of plagioclase-rich

rocks in the lower Martian crust, then it is insufficient to account for the observed density discrepancy (assuming a density of 2.7 g/cc for anorthosite and 3.1 g/cc for basaltic rock).

Several factors could contribute to resolving this apparent contradiction. First, the lower-crust spectrally mafic compositions detected in our study may contain a significant feldspar component that is not reflected in their spectral signatures due to the masking effect of mafic minerals (e.g., Extended Data Fig. 4). The THEMIS thermal infrared data suggest that these rocks follow a more gabbroic than ultramafic trend (Fig. 3), which could indicate the presence of troctolite or leucogabbro/leuconorite. Density measurements of leucogabbro/norite from the Bushveld Complex indicate such rocks are ~ 2.9 g/cc (Roelofse, 2010). However, determining whether such feldspar-rich gabbroic compositions are present requires further detailed analysis, and potentially additional data from future missions, such as higher spatial resolution orbiting spectrometers and imagers or landed missions that sample material uplifted from basin-forming impacts. Indeed, such rocks may have been observed already by the Perseverance rover at Jezero Crater (Bedford et al., 2024; Schmidt et al., 2025), which lies within the circum-Isidis region where Isidis uplift should be present.

Alternatively, additional factors such as impact induced fractures, alteration, or hydration must be invoked to explain the lower crustal density (Ehlmann et al., 2010; Bandfield et al., 2013; Sun and Milliken, 2015; Wordsworth et al., 2017; Goossens et al., 2017; Scheller and Ehlmann, 2020; Wieczorek et al., 2022; Carrasco et al., 2023). The observed association of plagioclase-rich outcrops with evidence for alteration in some locations supports the possibility that hydrothermal processes have affected the lower crust (e.g., Viviano and Phillips, 2019; Wu et al., 2025), potentially reducing its density through alteration of primary minerals and introduction of hydrated phases.

The igneous complexes suggested by our observations may also have implications for Mars' remnant crustal magnetism. Layered mafic intrusions and massif anorthosites on Earth often contain significant quantities of iron-oxide minerals such as magnetite and titanomagnetite, which can carry strong remanent magnetization. The shallow differentiated igneous complexes that may have formed during Mars' early history, when the planet had an active dynamo, could contain the minerals necessary to carry the strong crustal magnetic anomalies observed in the southern highlands.

Conclusions

Detection of widespread spectrally feldspathic outcrops within material uplifted by the Argyre basin-forming impact provides compelling evidence for the presence of feldspathic, possibly anorthositic, rocks as a component of Mars' lower crust. These spectral signatures indicate plagioclase dominance at the surface, but the bulk rock composition could range from anorthosite to leucogabbro/norite to gabbro. The light-toned appearance, presence of dark and light toned layering, color properties in HiRISE IRB data, higher reflectance than nearby mafic rocks in both HiRISE and CRISM data, and spatial association of these plagioclase-rich materials with olivine and pyroxene-bearing outcrops suggests they represent components of ancient, differentiated complexes of either igneous or impact origin (or both) that formed during Mars' early crustal development.

Observations of crustal compositions indicative of LMI-like complexes support models of early Martian evolution involving large-scale melting (Fig. 4), though the specific mechanism—whether mantle overturn, impact melting, or other processes—remains open to debate. The presence of similar plagioclase-rich materials uplifted by both the Argyre and Hellas impacts suggests their formation required a hemispheric or global process rather than isolated activity. This evidence provides constraints for refining theoretical and computational models of early Mars crust formation and offers insights into crustal differentiation processes on terrestrial planets lacking plate tectonics.

Missions targeting ancient crustal materials could provide additional constraints on their composition, age, and formation mechanisms. For example, detections of sulfide, spinel and oxide horizons in association with layered intrusive outcrops would support an interpretation that these rocks represent differentiated complexes. The Perseverance rover at Jezero crater has recently observed feldspar dominated (Bedford et al., 2024; Schmidt et al., 2025) and spinel-bearing rocks (Mansbach et al., 2024; Bedford et al., 2024; Lawson et al., 2025) that could represent material uplifted by the Isidis-forming impact, warranting further investigation.

Methods

Reflectance Spectroscopic Analyses

We used the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM; Murchie et al., 2007) onboard the Mars Reconnaissance Orbiter to identify plagioclase-rich outcrops in the circum-Argyre region. CRISM is a visible and near-infrared imaging spectrometer that provides high-resolution reflectance spectral data across a wavelength range of approximately 400 to 4000 nm. The CRISM data used in this study are the 72-band (410 – 3923 nm), 180 m/pixel Map-Projected Multispectral Reduced Data Record (MRDR) v4 data (Seelos et al., 2023; Murchie, 2024). The MRDR map tiles were analyzed using the Spectral Cube Analysis Tool (Phillips, 2024). Regions of interest were drawn using combinations of CRISM spectral parameters (Viviano et al., 2014) and spectra were then averaged from the ROIs and ratioed to in-scene spectrally bland regions – a common technique in CRISM analysis to remove scene-dependent instrument and atmospheric artifacts in the data. Interpretations of mineralogy present in each spectrum were made through visual comparison to reference spectral libraries (Minerals identified with CRISM Analysis [MICA], Viviano et al. 2014; USGS Spectral Library v7, Kokaly et al. 2017). Spectra were categorized based on the presence of diagnostic absorption features of the primary rock forming minerals olivine, pyroxene, and plagioclase (Fig. 3). It should be noted that the spectrally dominant phase used for categorical designations in this work is not always the volumetrically dominant phase in a rock. Therefore, categorical designations based on spectral features should not be confused with lithological interpretations (e.g., categorization as low-Ca pyroxene does not imply the rock is comprised of only orthopyroxene, but rather the reflectance spectrum is dominated by spectral signatures consistent with a low-Ca pyroxene).

Plagioclase was identified based on the presence of a broad absorption feature centered at approximately 1.25- μm (Fig. 3). Plagioclase spectra were differentiated from olivine spectra based on a reflectance peak near 900 nm that is present in plagioclase but absent in olivine. This diagnostic feature highlights the importance of using “joined” CRISM reflectance data products (that include both short and long detector channels) to identify plagioclase spectra. Pyroxene detections were categorized as low-calcium pyroxene (LCP) and high-calcium pyroxene (HCP) based on the minimum positions of the ~ 1 - and ~ 2 - μm absorption bands (Viviano et al., 2014). Where the pyroxene category presented as mixtures of olivine and pyroxene, pyroxene composition was determined solely by the presence and position of the 2- μm band. Olivine detections were categorized as “type 1” and “type 2” olivine (Ody et al., 2013) based on a k-means classification of 5 band parameters (band center, band depth, band asymmetry, slope between 1 and 1.3 μm , and full width half max) that quantify properties of the ~ 1 - μm olivine absorption feature (Extended Data Fig. 9). The complete python code and data to replicate the classification are attached to this article. Type 1 olivine is considered more forsteritic or finer grained and type 2 olivine spectra indicate either more fayalitic or coarser grain-size olivine; a coarse-grained low-iron olivine is spectrally similar to a fine-grained high-iron olivine (e.g., Mustard et al., 2005), which motivates the compositionally agnostic “type 1” and “type 2” designations (Ody et al., 2013). Spectra with a broad 1- μm olivine absorption and a broad 2- μm absorption consistent with HCP were categorized as olivine + HCP (Extended Data Fig. 4). Spectra were further categorized as Low, Moderate, or High confidence based on the strength and clarity of absorption features present on a qualitative basis (Fig. 3). Geologic units from Dohm et al. (2015) were used to assign each mineral unit with an associated geologic context: either uplifted by the Argyre impact (Argyre basin and rim materials, Argyre basin members) or uplifted by later large impacts (old and young crater materials) that excavated Argyre rim materials (Extended Data Table 3).

Thermal Emissivity Analyses

In addition to reflectance spectroscopy, 100 m/pixel, 9-band thermal emissivity data from the Thermal Emission Imaging System (THEMIS) onboard the 2001 Mars Odyssey orbiter (Christensen et al., 2004) were used to constrain the silica content of select spectrally feldspathic outcrops. THEMIS data with the warmest available surface temperatures were processed following the methods of Bandfield et al. (2004). In the absence of alteration, the wavelength of minimum emissivity is negatively correlated with the silica content of a rock (Lyon, 1965). Locations analyzed were ensured to indicate no evidence for alteration (hydration bands) in the co-located CRISM spectra. Felsic, mafic, and ultramafic rocks show minima at ~ 9.6 μm , ~ 10.5 μm , and 11 μm , respectively (Fig. 3). Notably, anorthositic compositions can display minima consistent with mafic/gabbroic compositions due to their similar silica contents (Fig. 3).

To assess the silica content of outcrops identified with CRISM, we calculated two metrics for the THEMIS emissivity minimum position: the cubic spline fit minimum (CSFM) and the weighted absorption center (WAC). The CSFM method fits a cubic spline to the THEMIS

emissivity data for bands 3 – 8 ($\lambda = 7.89 - 12.56 \mu\text{m}$) and returns the minimum of the cubic spline fit values (Rogers and Nekvasil, 2015). The WAC method calculates the wavelength value at which the cumulative absorption area equals half the total absorption area (Vincent and Thomson, 1972; Smith et al., 2013; Amador and Bandfield, 2016). The WAC metric has less variance than the CSFM and is more robust against idiosyncrasies in wavelength minimum values that may arise from a cubic spline fit; however, both measurements correlate inversely with the total silica content of the rock (Vincent and Thomson, 1972; Walter and Salisbury, 1989; Rogers and Nekvasil, 2015; Amador and Bandfield, 2016).

The relationship between CSFM and WAC can further elucidate the spectral character of rocks, particularly for anorthositic and gabbroic compositions compared to intermediate rocks. We calculated WAC and CSFM values for THEMIS-convolved reference spectra from the Arizona State University (ASU) spectral library (ASU, 2025), RELAB database (RELAB, 2025), Ward’s spectral library (Bryson, 2014), and from Dunn (2021). CSFM values show a wider range than WAC values (Fig. 3B). For felsic and intermediate rocks, CSFM values are lower than the corresponding WAC values for the same rock. For mafic, ultramafic, gabbroic, anorthositic, and pyroxenitic rocks the CSFM–WAC values show an approximate 1:1 relationship with some CSFM values being higher and some lower than the corresponding WAC values. For middling WAC values (~ 10.1 to $10.6 \mu\text{m}$) that could be interpreted as multiple rock types (e.g., intermediate, anorthositic, gabbroic), a relatively high CSFM value would indicate a more anorthositic or gabbroic rock (Fig. 3B).

Assessment of Color, Geomorphological and Thermophysical Properties

We used the 5 m/pixel Context Camera (CTX) global mosaic (Malin et al., 2007; Dickson et al., 2018) and 0.25 m/pixel HiRISE images (McEwen et al., 2010) to assess the color and texture of outcrops. For HiRISE IRB color analysis, dark subtraction methods were used after Tornabene et al. (2018) to correct for additive atmospheric residuals in the scenes that affect the spectral shape of the data. Even though efforts have been made to reduce spectral contributions from time-variable atmospheric dust aerosols (e.g., Ceamanos et al., 2013; Fernando et al., 2013), band ratios that utilize blue to green wavelengths will be influenced to some degree by the ubiquitous ferric dust on the surface of Mars; indeed, a BG/IR ratio (e.g., Extended Data Fig. 7) tends to be lower than expected compared to laboratory reference spectra of the same minerals on Earth (e.g., Delamere et al., 2010; Tornabene et al., 2018; Rangarajan et al., 2024). It should also be noted that grain size will also influence band ratios and other spectral parameters (e.g., Hapke, 1981; Clark and Roush, 1984). HiRISE scene ESP_031269_1395 overlaps with CRISM-identified plagioclase, LCP, and HCP + olivine outcrops. Regions of interest (ROIs) were drawn from areas overlapping the CRISM-detections to investigate the HiRISE spectral character and support interpretations of similarly colored outcrops in nearby HiRISE scenes (ESP_020799_1385, ESP_031269_1395) as representative of plagioclase-bearing or mafic-bearing compositions (LCP or olivine $\pm$ HCP).

Thermal inertia properties were examined using THEMIS global thermal inertia mosaic

(Ferguson et al., 2006; Edwards et al., 2009) and nighttime infrared data (Edwards et al., 2011) to distinguish between relatively consolidated and unconsolidated materials.

Impact Modeling

We employed the three-dimensional shock physics hydrocode iSALE3D (Amsden et al., 1980; Collins et al., 2004; Wünnemann et al., 2006) to simulate the impact event that formed the Argyre basin. Our simulations successfully reproduced a 1700 km diameter crater using a 280 km dunite projectile impacting at 12 km/s [the mean Martian impact velocity (Bottke et al., 1994)] and at an angle of 45° relative to the horizontal [the most probable impact geometry (Melosh, 1989)].

The Martian crust and mantle were represented using ANEOS equations of state for basalt (Pierazzo et al., 2005) and dunite (Benz et al., 1989), respectively. A pre-impact crustal thickness of 60 km was assumed, consistent with the expected range (Neumann et al., 2004; Wieczorek et al., 2022). The rheological behavior of the crust and mantle was modeled with a rock-like strength model (Collins et al., 2004), a damage model with exponential dependence on plastic strain (Collins et al., 2004; Johnson et al., 2016), a dilatancy model (Collins, 2014), and a thermal weakening model accounting for temperature- and pressure-dependent material softening (Ohnaka, 1995). The mantle was further treated with a viscoelastic-plastic rheology (Elbeshausen and Melosh, 2018). A comprehensive list of material parameters is provided in Extended Data Table 6.

The high-resolution zone of the computational domain comprised 360 cells in the horizontal direction, 114 cells vertically, and 180 cells in depth, comparable to recent three-dimensional simulations of the Chicxulub impact structure (Collins et al., 2020). To emphasize downrange ejecta transport, a horizontal offset of 48 cells was introduced.

Extended Data Figure 1 shows a top-down view of the distributions of the Lagrangian tracer particles ejected. These tracers track material as they move through the simulation. By color-coding particles at their final locations according to their depth of origin, we highlight the distribution of ejected stratigraphic units. For illustrative purposes and to highlight lower crustal and mantle material, we have plotted deeper tracers on top. The material in the down range direction is deposited both as ejecta and as central peak collapse flow aiding to the concentration of deeper material downrange.

The objective of this simulation was to investigate the general pattern of material excavation and deposition associated with an impact of Argyre’s scale. Future work should be conducted to refine and further constrain the impactor properties and target conditions.

Data availability

All data used in this study are available from public repositories. CRISM data are available from the Planetary Data System (PDS) Geosciences Node. THEMIS, CTX, and HiRISE data are available from their respective PDS nodes. Processed data products, including

interpreted spectral detection maps with associated metadata and THEMIS convolved reference spectra, as well as code used to produce the figures in this publication, are available at <https://doi.org/10.5281/zenodo.15596017>.

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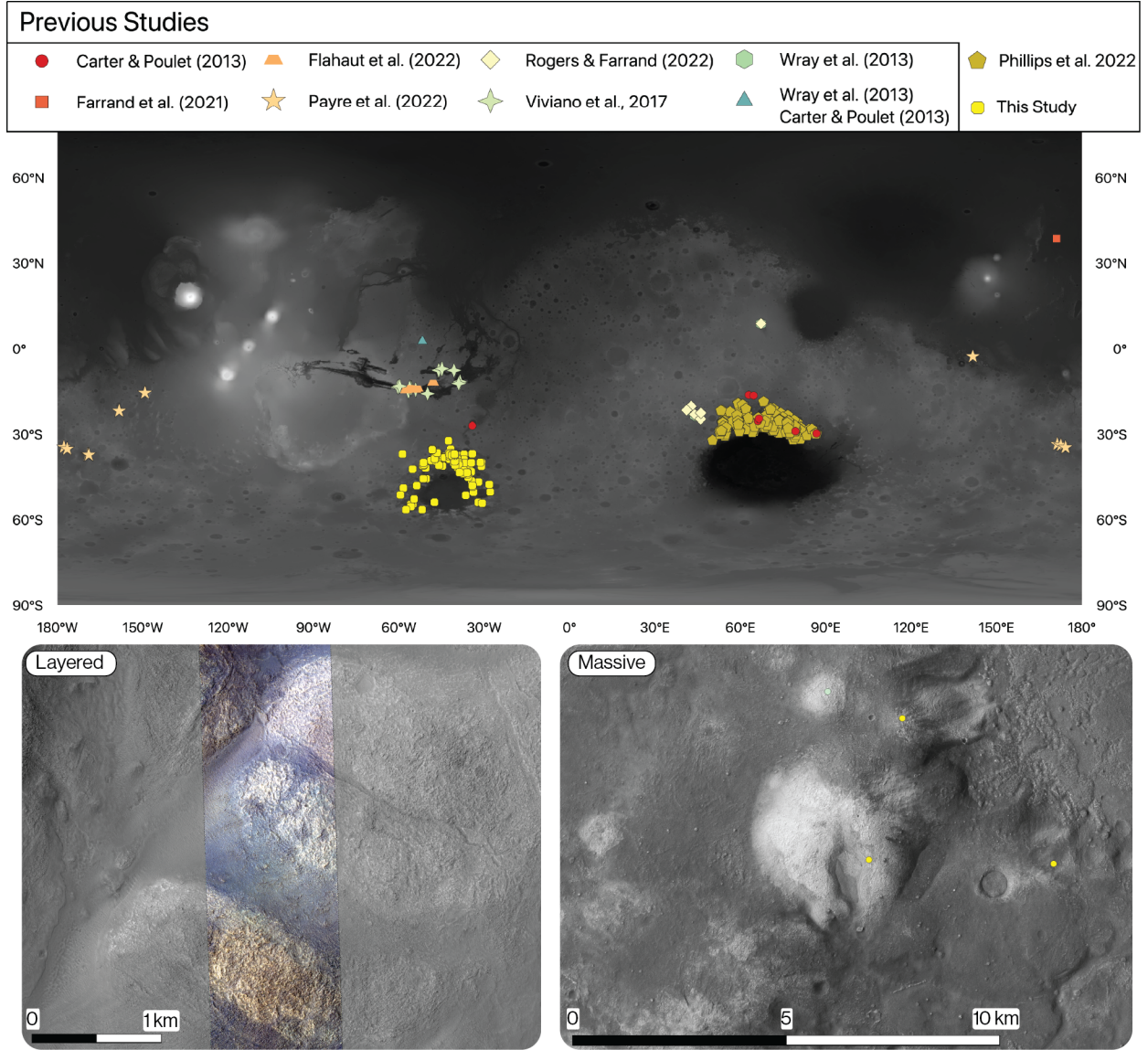


Figure 1: Top panel shows the global distribution of outcrops showing a plagioclase absorption feature in visible to near-infrared reflectance spectra. Results from this study in the circum-Argyre region are shown as yellow squares. Bottom left panel shows an example of possible layering between dark and light-toned units (HiRISE scene ESP_020799_1385). Bottom right panel shows an example light-toned massive textured outcrops with plagioclase reflectance signatures (yellow dots, mint green = alteration $\pm$ plagioclase) from the CTX global mosaic (Dickson et al., 2018).

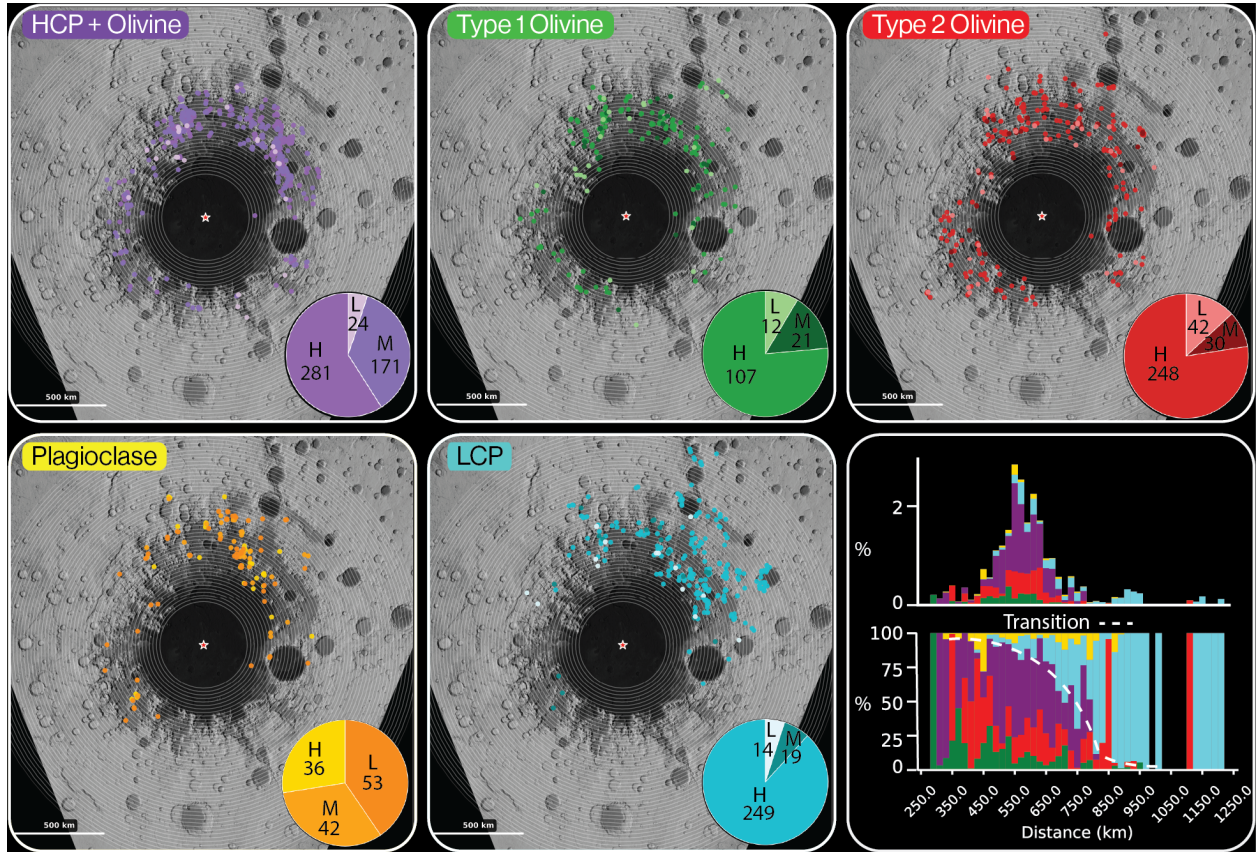


Figure 2: Detections of high-Ca pyroxene + olivine, type 1 and 2 olivine, plagioclase, and low-Ca pyroxene made with CRISM MRDR v4 data displayed on MOLA/HRSC shaded elevation base map (Ferguson et al., 2018). Proportions of low (L), moderate (M), and high (H) confidence detections are shown in the pie charts (c.f., Figure 3). White concentric circles on each map mark the boundaries of annular rings used for quantitative analysis, starting at 250 km radius and extending to 1250 km radius in 20 km increments (51 rings total). The bottom right panel shows the spatial distribution of each mineral category as a function of radial distance from the basin center, with each bar representing the area coverage within one 20 km-wide annular ring. The top histogram is normalized to the area of geological units from Dohm et al. (2015) investigated within each ring, and the bottom histogram is normalized by the total area of CRISM detections within each ring to elucidate relative trends in the data. The dashed line in the bottom histogram highlights the transition from olivine-dominated to LCP-dominated spectra. Quantitative data for the histograms are provided in Extended Data Tables 4 and 5, and a link to the code used to produce the plots can be found in the Data Availability section.

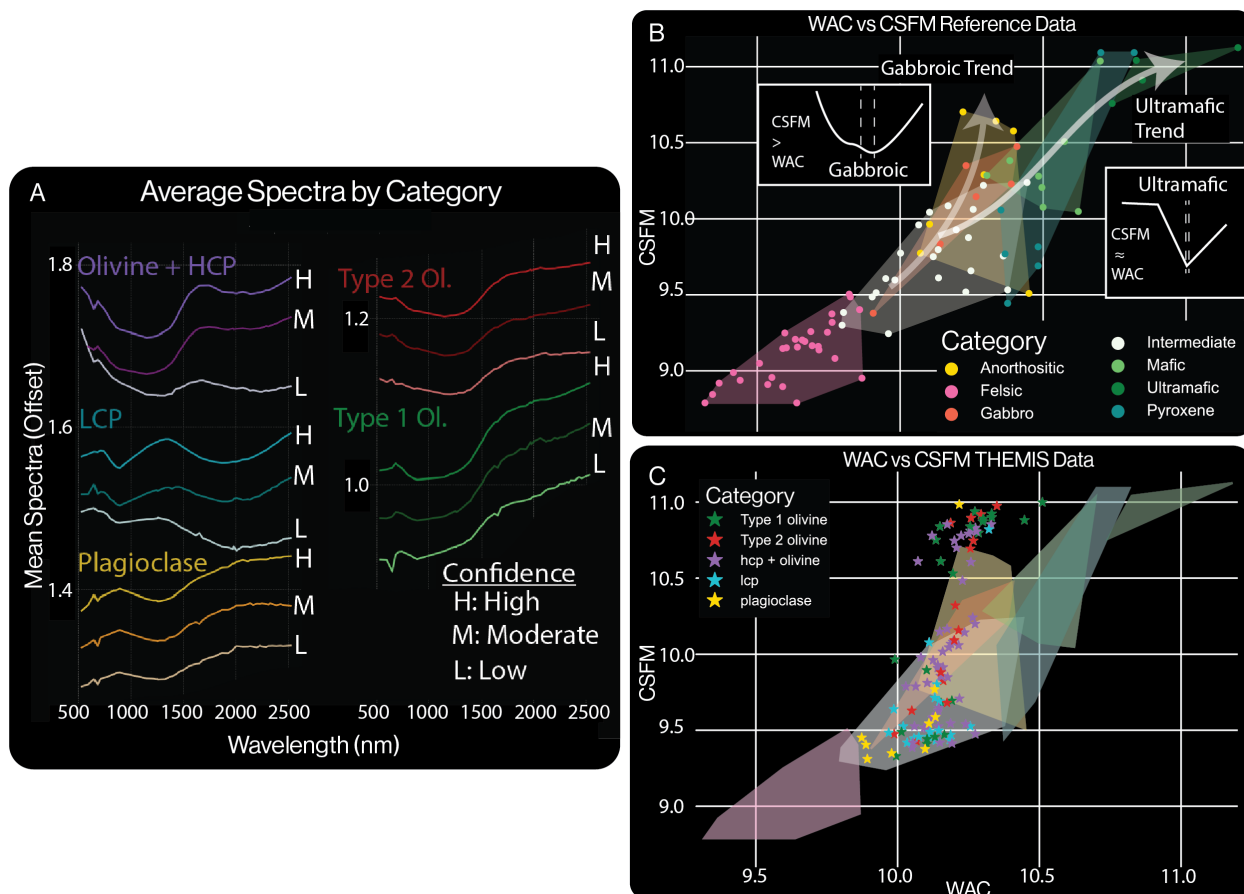


Figure 3: Reflectance and emissivity spectral character of outcrops surrounding Argyre. A) Average CRISM spectra (ratioed I/F, offset for clarity) for each category separated by low (L), moderate (M) and high (H) confidence based on absorption strength and clarity. B) Weighted absorption center (WAC) and cubic spline fit minimum (CSFM) values calculated on THEMIS-convolved reference library data (see methods for details) of felsic, intermediate, mafic, ultramafic, gabbroic, anorthositic, and pyroxenitic rocks. A minimum bounding polygon is drawn around each category type and displayed in panel C for reference. Anorthositic and gabbroic rocks show a trend of higher CSFM values compared to WAC values and pyroxenitic and ultramafic rocks tend to have closer to 1:1 correspondence between CSFM and WAC values. C) WAC and CSFM values calculated on ROIs detected with CRISM (THEMIS IDs: I09614002, I09639003, I17888007, I18025006) with regions from reference library spectra for interpretation. Outcrops show the highest overlap with intermediate, gabbroic, and anorthositic compositions.

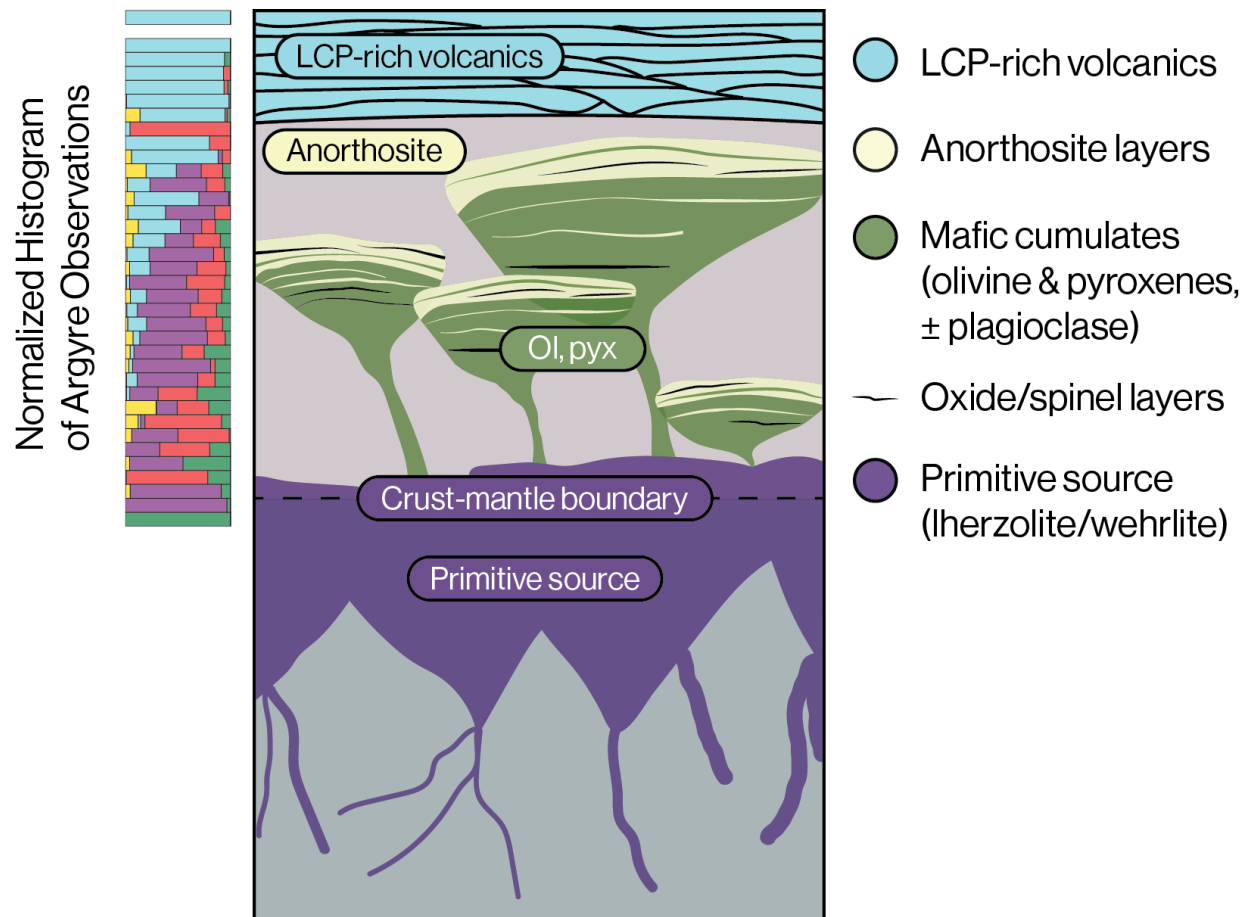


Figure 4: Conceptual model for the pre-Noachian crust exposed by the Argyre impact that could explain the observed stratigraphy at both regional (10s to 100s of kilometers) and local (1–5 kilometers) spatial scales. A primitive source of possible lherzolitic or wehrlitic composition produces basaltic melt extracts that undergo fractional crystallization at lower pressures to form layered-mafic-intrusion style complexes, likely containing oxide and spinel horizons. Low-Ca pyroxene rich volcanics form a secondary upper crust. Colors on the histogram are the same as Figure 2.